

Valuing global wildfire PM_{2.5} mortality caused by climate change

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Abstract

The social cost of carbon dioxide (SC-CO₂) estimates the monetized damages from an additional ton of CO₂ emissions and is influential in climate policy. Existing global damage estimates account for the effect of climate change on temperature-related mortality, agriculture, energy, and sea-level rise, yet omit the impact on wildfires. We address this gap by estimating the mortality damages attributable to fine particulate matter (PM_{2.5}) air pollution from climate-induced wildfires. To do so, we harmonize output from Earth System Models (ESMs) to derive country-specific relationships between global mean temperature (GMT) change and population-weighted PM_{2.5} exposure, and we parameterize a wildfire smoke mortality damage sector within the GIVE integrated assessment model. In Monte Carlo simulations that sample key damage-function parameters, we estimate a mean SC-CO₂ from wildfire PM_{2.5} mortality of \$47/tCO₂ (95% conditional interval: \$10.2–\$102.1). This raises the mean global total SC-CO₂ by 23%, from I \$208 to \$255 per ton CO₂ at a 2% discount rate. The regions contributing most to the global wildfire PM_{2.5} damage total are the Former Soviet Union, the United States, and Western Europe, driven by large estimated increases in population-weighted wildfire PM_{2.5} exposure per degree of warming. Since we account for only the excess mortality due to wildfire PM_{2.5}, we consider our estimates to be a lower bound of the SC-CO₂ attributable to climate-induced wildfires.

Long-term exposure to fine particulate matter air pollution ($\text{PM}_{2.5}$) is associated with increased all-cause mortality (Pope et al., 2020). Wildfire smoke is a significant and growing source of $\text{PM}_{2.5}$ (Park et al., 2024), and climate change is predicted to increase wildfires, smoke exposure, and associated mortality in many regions (Pierce et al., 2017; Zhao et al., 2025). However, the mortality damages from climate-induced wildfire smoke are missing from existing global estimates of the social cost of carbon dioxide (SC- CO_2). The Greenhouse Gas Impact Valuation Estimator (GIVE) integrated assessment model estimates the SC- CO_2 with four damage sectors—temperature-related mortality, agriculture, energy, and sea-level rise (Rennert et al., 2022), but it ignores climate-induced wildfire smoke. Extending GIVE, Qiu et al. (2026) estimates a partial SC- CO_2 accounting for wildfire $\text{PM}_{2.5}$ mortality in the United States, finding it to be a substantial damage component. No global estimate yet exists.

In this study, we develop a global damage function linking global mean temperature (GMT) change to wildfire smoke mortality, which we incorporate as a fifth damage sector in GIVE. To parameterize this function, we estimate country-level relationships between GMT change and population-weighted $\text{PM}_{2.5}$ exposure using data from five global gridded analyses of Earth Systems Models (ESMs).¹ Process-based ESMs explicitly represent vegetation growth, fuel loads, fuel combustibility, wildfires, and chemical transport. In contrast, Qiu et al. (2026, p. 5) apply two separate empirical models that link temperature to wildfire activity and wildfires to $\text{PM}_{2.5}$ concentrations, so their estimates do not account for the effects of past wildfires on future wildfire severity, intensity, or vegetation composition and species shifts that may substantially influence projected wildfire emissions, smoke, and associated mortality. The main advantage of our approach is that ESMs explicitly model those feedback dynamics on a global scale. The main disadvantage is that ESMs are computationally intensive, so global, gridded land-atmosphere-wildfire output is limited. We estimate the relationship between GMT and wildfire $\text{PM}_{2.5}$ exposure across 25 historical and future simulations. We combine these estimates with a concentration-response coefficient to compute mortality damages and a global SC- CO_2 (Section 3). Our analysis of ESM output makes our global damage function a useful complement to Qiu et al. (2026)’s empirical approach, and we estimate a similar partial SC- CO_2 attributable to wildfire smoke mortality in the US. Following Rennert et al. (2022), we assess uncertainty in our SC- CO_2 estimates via Monte Carlo simulation, sampling over both the concentration-response coefficient and the country-level GMT- $\text{PM}_{2.5}$ coefficients (Section 3.4). We discuss the strengths and limitations of our approach relative to Qiu et al. (2026) (Section 8) and compare our U.S. partial SC- CO_2 estimate to theirs (Section 9). In sensitivity tests, results from our preferred regression specification—using pooled data with model fixed-effects—are compared with results using data from individual ESM models (Section 10).

¹Pierce et al. (2017) use CESM, Park et al. (2024) use CLASSIC, SSiB4, and JULES, and Zhao et al. (2025) use CESM2, CESM2-WACCM, EC-Earth3-CC, and EC-Earth3Veg.

Our approach harnesses output from process-based models to characterize the relationship between GMT and wildfire PM_{2.5} exposure (Section 3.1). Process-based models explicitly represent physical mechanisms linking warming to vegetation growth, fuel loads, wildfire activity, and PM_{2.5} concentrations. This is important because the models account for future fuel availability under different warming and wildfire scenarios. Pierce et al. (2017, p. 4) use “a global dynamic vegetation model with carbon and nitrogen biogeochemistry, vertically resolved soil C, and nitrification/denitrification [CLM4.5BGC] to treat the response of vegetation to future climate and wildfires.” Park et al. (2024) “use wildfire-vegetation models [with] climate variables to dynamically simulate fuel load (biomass), fuel combustibility (soil moisture, relative humidity and air temperature), wildfire spread (wind speed) and fuel consumption (soil moisture).” Zhao et al. (2025) apply a machine learning to predict changes in vegetation biomass and future fuel consumption using output from process-based Earth System Models (ESMs)². The upshot is that Park et al. (2024) and Pierce et al. (2017) explicitly model feedbacks between wildfire activity, vegetation growth, and future fuel availability, while Zhao et al. (2025)’s projections capture climate- and land-use-driven variation in fuel availability by applying the vegetation biomass change ratio. These feedback dynamics are embedded in the Park et al. (2024) and Pierce et al. (2017) simulation output we use to estimate GMT-fire PM_{2.5} relationships. By contrast, Qiu et al. (2026, p. 5) apply two separate empirical models that link temperature to wildfire activity and wildfires to PM_{2.5} concentrations. Although they add a wildfire-fuel feedback to their projections, they acknowledge that their estimates “do not capture the effects of past wildfires on future wildfire severity, intensity, and shifts in vegetation composition or species, which may influence projected wildfire emissions, smoke, and mortality.” Our analysis of data from global process-based models makes our damage function a useful complement to the U.S.-focused empirical approach of Qiu et al. (2026), despite the limited data available from global, process-based models.

1 Results

We present results in two parts. First, we present GMT-PM_{2.5} relationships at the *grid-cell level*-. These estimates of the change in wildfire PM_{2.5} concentrations ($\mu\text{g}/\text{m}^3/\text{yr}$) per 1°C increase in GMT reveals where process-based models predict that warming most strongly increases wildfire PM_{2.5} exposure (Figure 1). Second, we report the global SC-CO₂ attributable to wildfire PM_{2.5} mortality using our GIVE wildfire PM_{2.5} damage function (Figure 2). This function uses *country-level* GMT-PM_{2.5} relationships—the change in population-weighted per-capita

²Zhao et al. (2025) use the term Earth System Model (ESMs), noting that the ESMs included are CESM2, CESM2-WACCM, EC-Earth3-CC, and EC-Earth3Veg. Pierce et al. (2017) use CESM (Section 4.1). In Supplementary Table 1, Zhao et al. (2025) confirm the land model component used but do not explicitly state whether the available wildfire components are active, which would explicitly capture feedbacks.

wildfire PM_{2.5} exposure ($\mu\text{g}/\text{m}^3/\text{yr}$) per 1°C increase in GMT—from our regression (Figure 3; Section 3.1).

In Figure 1, the strongest GMT-PM_{2.5} relationships are in boreal North America, boreal Asia, the Congo Basin, and the Amazon basin—regions³ where climate change has been shown to increase wildfire activity through rising temperatures and drying fuel loads (Park et al., 2024; Wimberly et al., 2024; Zhao et al., 2025). Moderate positive relationships are seen across temperate North America, Northern Africa, Europe, and parts of South and Southeast Asia. A minority of cells show near-zero or negative relationships in parts of Africa, South America, and Australia. In sub-Saharan Africa, there is evidence that competing mechanisms⁴ may offset one another as well as disagreement among ESMs on the impact of climate change in this region (Park et al., 2024). Parts of Australia shows unexpectedly flat or negative relationships, driven by flat or declining predicted wildfire PM_{2.5} in the input data from Zhao et al. (2025) and Pierce et al. (2017) (Figure 5).

Figure 2 reports the global SC-CO₂ attributable to wildfire PM_{2.5} mortality, with comparison to contributions from existing GIVE damage sectors. Adding to the uncertainty already captured in GIVE, we incorporate conditional uncertainty in the wildfire PM_{2.5} damage function parameters (Section 3.4). The wildfire PM_{2.5} SC-CO₂ has a mean of \$47/ton CO₂ (95% conditional interval: \$10.2–\$102.1) at a 2% discount rate (Panel A). Panel B shows that wildfire PM_{2.5} mortality represents the third-largest damage sector after temperature-related mortality (\$96.1) and agriculture (\$99.8), increasing the total global SC-CO₂ by 23%, from \$208 to \$255. Our U.S. partial SC-CO₂ from wildfire PM_{2.5} is somewhat smaller to that found by Qiu et al. (2026) (Section 9). Our regional decomposition reveals substantial regional heterogeneity in SC-CO₂ estimates (Table 1).

The SC-CO₂ results are driven by the *marginal* excess death associated with wildfire PM_{2.5}, computed by subtracting projected excess death in a baseline model from the same projection with one additional ton of CO₂, where excess death is a linear function of GMT change (Equation (2)). Accordingly, warming induces an average of 264,844 excess deaths per year globally at 1.5°C, rising to 528,136 at 3°C and 651,380 at 4°C. We compare our U.S. excess death estimates and damage function shape to Qiu et al. (2026) in Section 9. Relative to the Qiu et al. (2026) quadratic, our linear function projects more deaths at low warming levels and fewer deaths at high warming levels. However, on average, the Qiu et al. (2026) quadratic yields higher *marginal* excess deaths across most of the observed temperature range—GMT change > 1.7°C—which

³Climate change is expected to increase wildfire activity in large parts of Russia and Canada (high-latitude boreal forests areas) driven by warming and increasing drought (Zhao et al., 2025). Wildfire activity in the Congo Basin tropical forests has also increased, with year-to-year variation synchronized with rising temperatures and vapor pressure deficit, alongside deforestation (Wimberly et al., 2024).

⁴Higher temperatures may increase ignition potential while also driving evapotranspiration that may reduce grass productivity and fuel loads (Park et al., 2024).

drives the SC-CO₂ results.

Grid cell estimates: Change in fire PM_{2.5} concentration ($\mu\text{g}/\text{m}^3$) per 1°C GMT increase

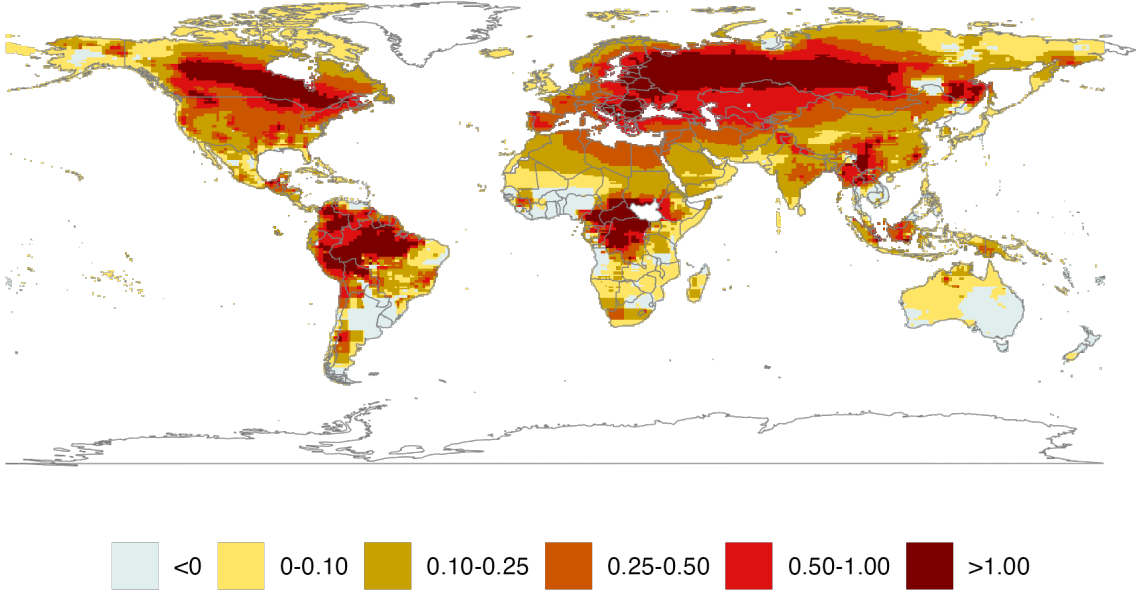


Figure 1: Grid-cell estimates of β_i , the change in wildfire PM_{2.5} concentration ($\mu\text{g}/\text{m}^3/\text{yr}$) per 1°C increase in GMT, estimated from a grid-cell-level fixed-effects regression pooling data from Park et al. (2024), Pierce et al. (2017), and Zhao et al. (2025).

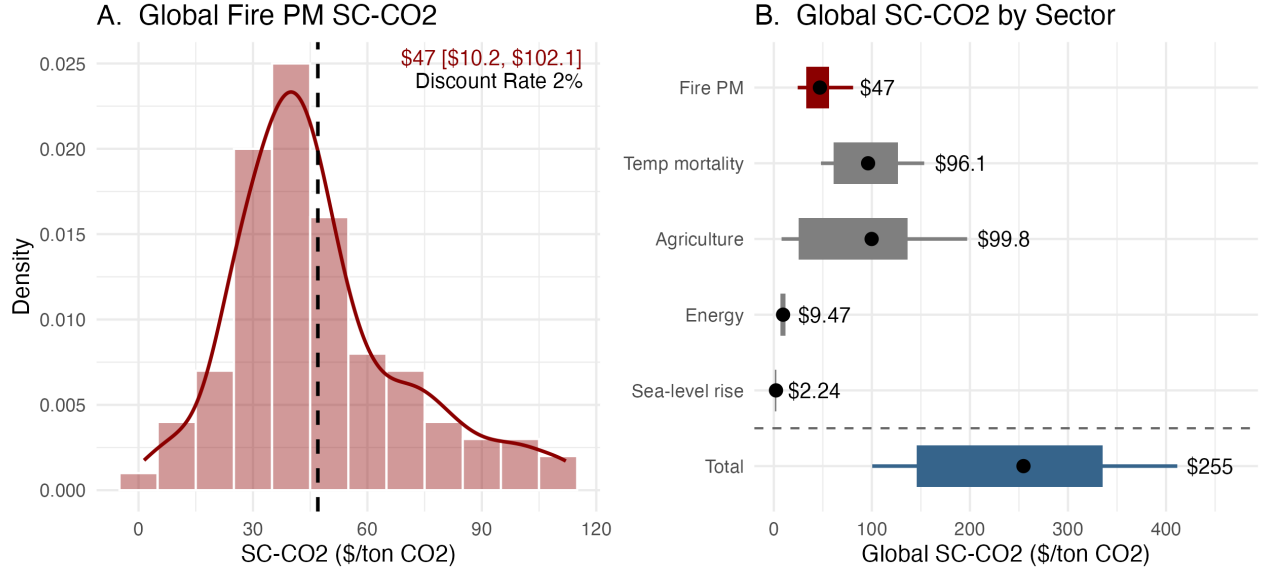


Figure 2: Global SC-CO₂ estimates at a 2% discount rate from 100 Monte Carlo simulations. (A) Distribution of the wildfire PM_{2.5} SC-CO₂ component; the dashed line marks the mean (\$47/ton CO₂; 95% conditional interval: \$10.2–\$102.1). (B) SC-CO₂ by damage sector; the dot marks the mean, the box spans the 25th–75th percentile, and whiskers span the 10th–90th percentile.

2 Discussion

Our key contribution is to estimate a global SC-CO₂ incorporating damages from wildfire PM_{2.5} mortality, which no prior studies have provided. We estimate that adding this fifth damage sector increases the global SC-CO₂ by 23%, from \$208 to \$255 per ton CO₂ in the GIVE integrated assessment model. At \$47/ton CO₂, wildfire PM_{2.5} mortality is the third-largest damage sector in GIVE, representing a substantial source of damages missing from existing global SC-CO₂ estimates. Our global study complements Qiu et al. (2026)’s finding that the partial U.S. SC-CO₂ from wildfire PM_{2.5} mortality is \$11.2. This suggests that current estimates of the SC-CO₂ guiding policymakers likely understate the cost of greenhouse gas emissions and the benefit of climate mitigation.

Our estimates are subject to several limitations. First, the concentration-response coefficient is drawn from a meta-analysis of long-term all-source PM_{2.5} exposure in Europe, North America, and Asia (Pope et al., 2020), and is applied uniformly across all countries, abstracting from regional heterogeneity in population susceptibility and background pollution levels. Second, the GMT-wildfire PM_{2.5} slope is identified from only 25 observations per country spanning five process-based wildfire PM_{2.5} models. This allows us to estimate a linear relationship between GMT and PM_{2.5} exposure, but it does not allow us to test for curvature or allow the relationship to change over time. Third, our Monte Carlo intervals are conditional on the linear functional

form, the set of source studies, and the assumption that model fixed effects absorb cross-model differences in levels rather than slopes. Our analysis cannot capture the full structural uncertainty in the GMT-PM_{2.5} relationship (Section 3.4). We discuss these limitations in detail in Sections 3.4 and 8.

Finally, this paper quantifies just one cost of climate-induced wildfires: PM_{2.5} mortality. Wildfire smoke also contributes to non-fatal cardiovascular and respiratory disease (Johnston et al. 2021), and people are willing to pay to avoid these health impacts (Jones 2018). Wildfire smoke also sharply reduces labor market earnings (Borgschulte et al. 2024). Beyond PM_{2.5} emissions, wildfires emit substantial greenhouse gas emissions (Byrne et al. 2024), so wildfires may create climate tipping points (Davis et al. 2019). Moreover, wildfires damage homes and reduce property values Bayham et al., 2022. It is difficult to quantify all the costs of wildfires, but in the United States, research suggests that the total cost of wildfires is at least 1.7- 2.5 times the cost of wildfire smoke mortality (Thomas et al. 2017).⁵ As a result, our estimates provide a lower-bound estimate of wildfires’ impact on the social cost of carbon.

3 Methods

3.1 GMT change and wildfire smoke exposure

Together, Pierce et al. (2017), Park et al. (2024), and Zhao et al. (2025) provide globally gridded data for 25 simulations of long-term wildfire PM_{2.5} exposure. Park et al. (2024) models climate-induced wildfire smoke for six decades from the 1960s to 2010s across three ESMs: Classic, Jules, and SSib4. Pierce et al. (2017) models wildfire smoke in the 2040s and 2090s along two climate pathways using CESM. Zhao et al. (2025) models wildfire smoke in the 2090s along two climate pathways. We combine these data to estimate the country-level relationship between wildfire PM_{2.5} exposure and global mean temperature (GMT) change.⁶ We estimate this regression for

⁵Thomas et al. review the literature on wildfire costs, which they summarize in Table 4.3. According to their low estimates, excess deaths from PM_{2.5} mortality account for 40% of total costs; according to their high estimates, they account for 58% of total costs. Thomas et al. (2017) do not consider the effect of smoke on productivity or the cost of greenhouse gas emissions.

⁶We interpret GMT as a policy-relevant index of global climate change rather than as the direct physical driver of wildfire PM_{2.5}. wildfire activity and smoke exposure respond mechanistically to regional temperature, vapor pressure deficit, precipitation, fuel availability, vegetation, ignition, suppression, land use, and atmospheric transport (Park et al., 2024; Pierce et al., 2017; Wimberly et al., 2024; Zhao et al., 2025). Therefore, β_c should not be interpreted as the causal effect of GMT on wildfire PM_{2.5}, but as a reduced-form mapping from GMT to the wildfire-PM_{2.5} exposure implied by the available model ensemble. This parameterization is appropriate given the limited availability of globally harmonized wildfire-PM_{2.5} projections and allows us to construct a country-level damage-function parameter for GIVE, where GMT is the state variable linking emissions to climate damages.

each country c :

$$\bar{P}M_{ctsm} = \beta_c T_{ts} + \alpha_{cm} + \varepsilon_{ctsm}, \quad (1)$$

where $\bar{P}M_{ctsm}$ is population-weighted⁷ per-capita wildfire PM_{2.5} exposure ($\mu\text{g}/\text{m}^3/\text{yr}$) for country c , at time period t , in climate scenario s , and for wildfire model m ; T_{ts} is the GMT change relative to the 1850-1900 pre-industrial baseline ($^{\circ}\text{C}$); α_{cm} is a country-fire-model intercept to control for across model differences;⁸ and ε_{ctsm} is a residual. The coefficient of interest, β_c , captures the change in population-weighted per-capita wildfire PM_{2.5} exposure associated with a 1°C increase in GMT for country c . We assume that this relationship is linear and constant over time⁹.

We estimate equation (1) at two spatial scales. At the country level, the dependent variable is population-weighted per-capita wildfire PM_{2.5} exposure $\bar{P}M_{ctsm}$, yielding the country-specific coefficient β_c that enters the GIVE damage function. At the grid-cell level, the dependent variable is grid-cell wildfire PM_{2.5} exposure PM_{itsm} , yielding a grid-cell-specific coefficient β_i —the estimated change in wildfire PM_{2.5} concentration per 1°C increase in GMT at grid cell i —used solely to characterize the spatial pattern of the GMT-fire PM_{2.5} relation (Figure 1).

The dependent variable is constructed by combining three sources of gridded wildfire PM_{2.5} data (Section 4.1), each paired with its corresponding GMT value. Park et al. (2024) provide historical decadal mean wildfire PM_{2.5} for the 1960s through 2010s under three wildfire models (CLASSIC, JULES, SSiB4), yielding 18 observations per country (3 models $\times$ 6 decades). Pierce et al. (2017) provide projections for a baseline period (2001–2010) and two future periods (2041–2050, 2091–2100) under RCP4.5 and RCP8.5 using the CESM wildfire model, contributing 5 observations. Zhao et al. (2025) provide two additional projections for 2095–2099 under SSP2-4.5 and SSP5-8.5, for a total of 25 observations per country. These gridded wildfire PM_{2.5} exposure data are aggregated to population-weighted per-capita country-level exposures.

The five wildfire models differ systematically in their absolute estimates of PM_{2.5} exposure

⁷See Section 4.3 for population-weighting detail.

⁸We assume each model has general assumptions that lead to level differences in GMT-wildfire smoke relationships. Our approach controls for these differences and uses within-model variation to evaluate the relationship between GMT and wildfire smoke.

⁹Qiu et al. (2026) estimate a quadratic damage function relating GMST to wildfire-smoke mortality rates in the United States. A more flexible functional form is appropriate in their setting because their analysis is U.S.-specific and uses a much richer simulation design: they combine an empirically estimated North American climate-fire-smoke model with projections from 28 CMIP6 climate models, yielding 784 climate simulations spanning approximately $0.95\text{--}6.93^{\circ}\text{C}$ of warming. By contrast, our objective is to estimate country-specific GMT-fire PM_{2.5} exposure relationships globally using harmonized output from a small number of heterogeneous wildfire PM_{2.5} modeling studies. For most countries, this provides relatively few distinct warming levels and limited within-model variation with which to identify curvature. We therefore impose a linear specification as a parsimonious approximation to the GMT-fire PM_{2.5} relationship. This avoids fitting curvature that is weakly identified by our data and avoids imposing stronger assumptions about high-warming extrapolation. In this context, the linear specification is also conservative relative to convex specifications.

due to differences in model structure rather than the GMT-fire relationship of interest in this paper. Without accounting for these differences, we would risk attributing to GMT variation in exposure levels that instead reflects differences across wildfire models. We address this by including model fixed effects α_{cm} , which absorb model-specific mean exposure differences so that β_c is identified from within-model covariation between GMT and wildfire PM_{2.5} across periods and scenarios.¹⁰ This specification estimates a pooled within-model slope, β_c , across the five wildfire models. If wildfire models differ not only in their mean exposure levels but also in their GMT-fire PM_{2.5} slopes, then β_c should be interpreted as an average model-implied slope. Conventional regression standard errors capture residual variation around this pooled slope, but they do not separately identify between-model slope heterogeneity as a distinct source of structural uncertainty (Section 3.4).

The resulting β_c estimate is used in the GIVE integrated assessment model as the country-level damage function parameter linking GMT change to wildfire PM_{2.5} exposure and its associated mortality. Across 174 countries, β_c is positive for 86% of countries (median = 0.18 $\mu\text{g}/\text{m}^3/\text{yr}$ per $^\circ\text{C}$), indicating that warming broadly increases population exposure to wildfire PM_{2.5}. Estimates are statistically significant at the 5% level for 60% of countries. The largest effects are concentrated in northern North and South America, northern Asia, central Africa, and parts of Southeast Asia, while fewer countries—predominantly in Africa and South America, as well as Australia—have negative β_c (Figure 3). For Australia, negative relationships at the grid-cell level in the most populous regions of the country (Figure 1) translate into declining population-weighted exposure with increasing GMT.

¹⁰Figure 5 in the Appendix illustrates this for select countries.

β_c across countries: Change in per-capita fire PM_{2.5} ($\mu\text{g}/\text{m}^3/\text{yr}$) per 1°C GMT increase

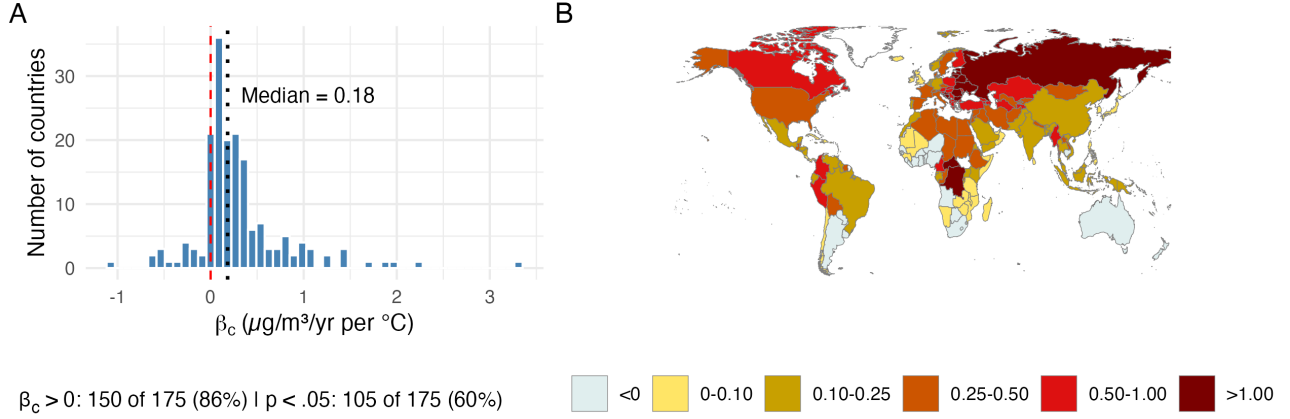


Figure 3: Country-level estimates of β_c , the change in population-weighted per-capita wildfire PM_{2.5} exposure ($\mu\text{g}/\text{m}^3/\text{yr}$) per 1°C increase in GMT. (A) Distribution of β_c across 175 countries; the dashed red line marks zero and the dotted black line marks the median ($0.18 \mu\text{g}/\text{m}^3/\text{yr}$ per °C). (B) Heat map of β_c by country.

3.2 PM_{2.5} exposure and mortality

Defining a wildfire PM_{2.5} damage function in GIVE requires a parameter relating PM_{2.5} exposure and mortality. Following Qiu et al. (2025) and Qiu et al. (2026)¹¹, we source this parameter from the Pope et al. (2020) meta-analysis of 33 selected studies in Europe, North America, and Asia of long-term exposure to PM_{2.5}. Using a random effects model, they estimate a hazard ratio (HR) of 1.08 (95% CI = 1.06–1.11) for all-cause mortality per $10 \mu\text{g}/\text{m}^3$. This estimate is a bit lower than the Orellano et al. (2024) meta-analysis, which reports a central estimate of 1.095 (95% CI: 1.064–1.127) per $10 \mu\text{g}/\text{m}^3$. We use the Pope et al. (2020) estimate to calculate the percent increase in all-cause mortality rates per $\mu\text{g}/\text{m}^3$ as $\gamma = \frac{1.08-1}{10} = 0.008$. This 0.8% increase per $\mu\text{g}/\text{m}^3$ is applied to all countries globally.

3.3 GIVE wildfire PM_{2.5} Damage Function

Our GIVE damage function takes a linear form analogous to the existing temperature-related mortality module in Rennert et al. (2022). We translate warming-induced wildfire PM_{2.5} exposure into monetized mortality damages similarly. For country c at time t , monetized damages

¹¹These studies use the all-source PM_{2.5} Pope et al. (2020) result for exposure levels outside historical observations in the U.S., those $> 6 \mu\text{g}/\text{m}^3$, while using their own wildfire PM_{2.5}-mortality response estimate for exposure $< 6 \mu\text{g}/\text{m}^3$. We instead apply the Pope et al. (2020) result for all exposure levels. Since there is evidence of greater toxicity from wildfire PM_{2.5} (Qiu et al., 2025), we believe our estimated impact to be conservative.

are

$$C_{ct} = V_{ct} \cdot M_{ct} \cdot \gamma \cdot \beta_c \cdot T_t, \quad (2)$$

where V_{ct} is the value of a statistical life (VSL) in USD, M_{ct} is the baseline all-cause mortality count, $\gamma = 0.008$ is the concentration-response coefficient (the proportional increase in mortality risk per $1 \mu\text{g}/\text{m}^3$ increase in long-term $\text{PM}_{2.5}$ exposure), β_c is the country-specific regression coefficient estimated in equation (1) (the change in population-weighted per-capita wildfire $\text{PM}_{2.5}$ exposure per 1°C increase in GMT), and T_t is the GMT change relative to the pre-industrial baseline ($^\circ\text{C}$).

The product $M_{ct} \cdot \gamma \cdot \beta_c \cdot T_t$ gives the number of excess deaths attributable to warming-induced wildfire $\text{PM}_{2.5}$ in country c at time t . Multiplying by V_{ct} converts these excess deaths to monetary damages. The VSL is sourced from (Rennert et al., 2022): a baseline U.S. VSL for 2020 is adjusted for each country and year using GDP per capita with an income elasticity of one. The baseline all-cause mortality count, M_{ct} , is likewise sourced from (Rennert et al., 2022), which projects these values for country and year from 2025 to 2300.

3.4 Parameter sampling and sensitivity analysis

Following Rennert et al. (2022), GIVE propagates uncertainty through Monte Carlo simulation. In the baseline GIVE model, each simulation samples uncertain socioeconomic and emissions trajectories (RFF-SPs), climate model parameters (FaIR), sea-level model parameters (BRICK), and damage-function coefficients. We retain this existing GIVE uncertainty structure and add parameter sampling for the two quantities specific to the wildfire $\text{PM}_{2.5}$ damage sector: the concentration-response coefficient γ and the country-level GMT-exposure coefficient β_c .

For γ , we draw from a normal distribution with mean $\bar{\gamma} = 0.008$ and standard error $SE(\gamma) = 0.00128$, derived from the 95% confidence interval of the Pope et al. (2020) hazard ratio ($\text{HR} = 1.08$; 95% CI: 1.06–1.11 per $10 \mu\text{g}/\text{m}^3$), converted to a per- $\mu\text{g}/\text{m}^3$ mortality response as $\gamma = (\text{HR} - 1)/10$.¹² For β_c , we draw independently for each country from a normal distribution with mean $\hat{\beta}_c$ and standard error $SE(\hat{\beta}_c)$ from the country-level fixed-effects regression in equation 1. These draws generate a distribution of wildfire $\text{PM}_{2.5}$ mortality damages and associated SC- CO_2 estimates conditional on our preferred damage-function specification.

These simulations should be interpreted as a parameter-sensitivity analysis rather than a full characterization of uncertainty. The resulting intervals reflect uncertainty in the concentration-response coefficient and sampling uncertainty in the estimated country-level GMT–wildfire $\text{PM}_{2.5}$ coefficients, conditional on the source studies used, the harmonization procedure, the linear GMT-exposure specification, the fixed-effects pooling structure, and the global transfer

¹²Specifically, $SE(\gamma) = (\gamma_{upper} - \gamma_{lower}) / (2 \times 1.96) = (0.011 - 0.006) / 3.92 \approx 0.00128$.

of the $\text{PM}_{2.5}$ mortality coefficient. They do not fully propagate uncertainty from alternative wildfire models, between-model heterogeneity in GMT–wildfire $\text{PM}_{2.5}$ slopes, alternative functional forms, regridding choices, regional heterogeneity in smoke toxicity or population vulnerability, future adaptation, or changes in within-country population exposure. We therefore report these intervals as conditional Monte Carlo sensitivity intervals, not as full structural uncertainty bounds.

To evaluate the sensitivity of the results to model structure, we also compare the preferred pooled fixed-effects specification to estimates using individual wildfire models where feasible. This comparison is reported separately from the Monte Carlo parameter sampling because the small number of source models does not support a formal decomposition of between-model structural uncertainty. Together, the Monte Carlo simulations and model-specific comparisons indicate how the wildfire $\text{PM}_{2.5}$ SC- CO_2 varies under our maintained assumptions, while also highlighting the sources of uncertainty that remain outside the reported intervals.

4 Appendix: Data

4.1 Fire PM_{2.5} Source Data and Regridding

Pierce et al. (2017), Park et al. (2024), and Zhao et al. (2025) each generate global gridded wildfire PM_{2.5} datasets and project how climate change alters wildfire activity and associated concentrations, but differ in their temporal coverage, methodology, and treatment of non-climate drivers. All studies simulate gridded emissions estimates from wildfire, which are fed into chemical transport models to produce wildfire PM_{2.5} contributions to total PM_{2.5}. The wildfire PM_{2.5} contributions are isolated as the difference between simulations with and without wildfire emissions conducted under the same scenario. GMT change is a key driver of wildfire PM_{2.5} outcomes in all studies. However, another key factor that varies in all studies is land cover and land use (LCLU), a variable capturing deforestation. Although Pierce et al. (2017, p. 11) state it is “obvious that future wildfire pollution will be primarily driven by climate,” LCLU changes also impact outcomes.

Pierce et al. (2017) use Community Earth System Model (CESM)—a coupled chemistry-climate-land model—to simulate wildfire activity and PM_{2.5} concentrations for a baseline period (2001–2010) and future periods (2041–2050 and 2091–2100) under RCP4.5 and RCP8.5. Their simulation design isolates the effect of climate change on wildfire activity by including both scenarios in which population density varies with SSP trajectories and scenarios in which it is held constant at year-2000 levels. In the chemical transport step, anthropogenic and biogenic emissions are likewise held constant within each snapshot period. We use only the population-constant simulations, as our regression targets the GMT-fire PM_{2.5} relationship excluding the impact of population change (Section 3.1).

Park et al. (2024) simulate historical wildfire PM_{2.5} from the 1960s through the 2010s using three wildfire-vegetation models (CLASSIC, SSiB4, and JULES) driven by historically observed climate forcings. In the chemical transport model, anthropogenic emissions are set to observed historical values for each decade from Community Emissions Data System Inventory (CEDS) and meteorological conditions are fixed at a single reference year, to attribute to climate change the variation in wildfire PM_{2.5} across decades.¹³ Their climate change attribution involves factual and counterfactual simulations. For consistency with the Pierce and Zhao datasets, we use the factual simulations, which span six decadal means across three wildfire models, yielding 18 observations per country.

Zhao et al. (2025) develop a machine learning model of burned area trained on Global wildfire Emissions Database version 4.1s (GFEDv4.1s) observed burned area, ERA5 climate reanalysis, and Earth System Model (ESM) estimates. The Coupled Model Intercomparison

¹³Specifically, Park et al. (2024) fix GEOS-Chem meteorological conditions at 2016 values for all simulation periods (Table 1 of Park et al. (2024)).

Project Phase 6 (CMIP6)-projected climate variables are used to predict future global wildfire gridded emissions estimates for 2095–2099 under SSP2-4.5 and SSP5-8.5. These emissions enter into the GEOS-Chem chemical transport model to produce wildfire PM_{2.5} concentrations. Land-use inputs follow scenario-specific Land Use Harmonization 2 (LUH2) trajectories, so land use varies with the warming scenario. Anthropogenic emissions likewise vary with the SSP scenario.

We regrid PM_{2.5} data from these multiple sources and native resolutions onto a unified $0.5^\circ \times 0.5^\circ$ global grid (720×360), and subsequently merge the gridded fields with population data using longitude–latitude matching. The only difference across the three scripts lies in the resampling method: the Park et al. (2024) dataset is interpolated using bilinear interpolation, the Pierce et al. (2017) dataset is remapped using nearest-neighbor matching, and the Zhao et al. (2025) dataset is aggregated using area-weighted averaging. All other processing steps are identical, including the definition of the target grid, coordinate harmonization, and the final left join with population data based on exact longitude-latitude pairs. The resulting grid is defined at the center of each 0.5° cell, with longitude spanning -179.75° to 179.75° and latitude spanning -89.75° to 89.75° , at 0.5° intervals.

4.2 Global Mean Temperature Change

Global mean temperature change values relative to pre-industrial times are associated with each PM_{2.5} simulation. For the Park et al. (2024) simulations, we assign historically observed GMT changes (OWID, 2026). For Zhao et al. (2025), we assign GMT changes projected under SSP2-4.5 and SSP5-8.5 (Anthoff, 2026). For Pierce et al. (2017), we use their gridded GMT change data relative to 2005, to which we apply a pre-industrial offset (OWID, 2026).

4.3 Appendix: Population Weighting

For grid cell i in country c , let PM_{itsm} denote wildfire PM_{2.5} exposure ($\mu\text{g}/\text{m}^3/\text{yr}$) under time period t , scenario s , and wildfire model m . Each observation is already a period average: a decade mean for Park et al. (2024) and Pierce et al. (2017), and a five-year mean for Zhao et al. (2025).

Baseline population $p\bar{op}_{ic}$ is the average annual population in cell i of country c over 2001–2010, sourced from Pierce et al. (2017). Population is held fixed at this baseline across all periods and scenarios to isolate changes in wildfire PM_{2.5} concentrations. Country-level baseline population is $p\bar{op}_c = \sum_i p\bar{op}_{ic}$.

Population-weighted per-capita wildfire PM_{2.5} exposure for country c , period t , scenario s ,

and wildfire model m —the dependent variable in equation (1)—is:

$$P\bar{M}_{ctsm} = \frac{\sum_i PM_{itsm} \cdot p\bar{o}p_{ic}}{p\bar{o}p_c}.$$

5 Appendix: Regression Descriptive Statistics

Figure 4 summarizes the distribution of the two regression inputs across the 4,350 country-period-scenario-model observations used in equation (1). Panel A shows the cross-country distribution of population-weighted per-capita wildfire $PM_{2.5}$ exposure ($\mu g/m^3/yr$) by wildfire model. Pooled observations across all models, the distribution is heavily right-skewed: the median is 0.67 while the mean is 1.92, and the 90th percentile (4.75) is more than six times the 10th percentile (0.08). Mean exposure differs substantially across wildfire models—ranging from 1.07 (JULES) to 2.53 (CLASSIC)—motivating the inclusion of wildfire-model fixed effects to absorb these level differences before estimating the GMT change-exposure slope. CESM exposure levels are generally higher than the Park and Zhao models across the distribution: the CESM 25th percentile (0.77) exceeds the median of all three Park models (0.35–0.55) and the Zhao median (0.64).

Panel B shows the discrete GMT change values ($^{\circ}C$ relative to the pre-industrial) assigned to each data source. The six Park et al. (2024) historical decade means span 0.25–1.09 $^{\circ}C$, the low end of the temperature range. The five Pierce et al. (2017) projections extend from 1.04 $^{\circ}C$ (baseline period) to 4.47 $^{\circ}C$ (RCP8.5, 2091–2100), and the two Zhao et al. (2025) projections contribute values of 2.92 $^{\circ}C$ (SSP2-4.5) and 4.96 $^{\circ}C$ (SSP5-8.5). Together the three sources span approximately 0.25–5.0 $^{\circ}C$, enabling identification of β_c across a wide range of warming levels.

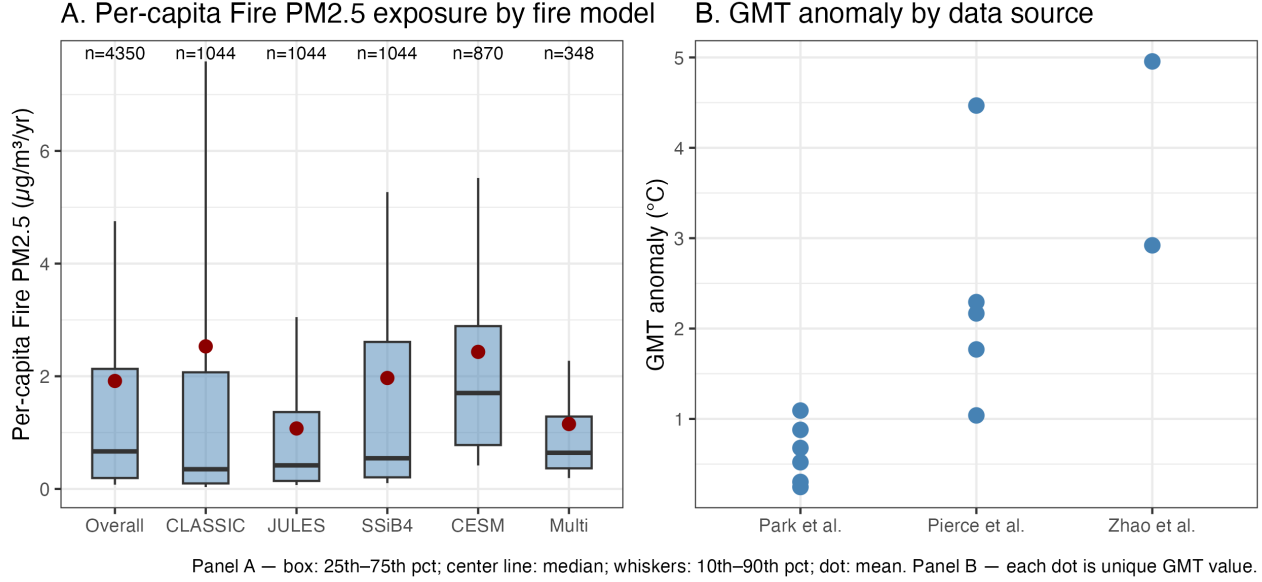


Figure 4: Distribution of regression inputs. (A) Cross-country distribution of population-weighted per-capita wildfire $\text{PM}_{2.5}$ exposure ($\mu\text{g}/\text{m}^3/\text{yr}$) for the full sample and by wildfire model; box spans the 25th–75th percentile, center line is the median, whiskers span the 10th–90th percentile, and the red dot marks the mean. (B) Discrete GMT change values ($^{\circ}\text{C}$ relative to the 1850–1900 pre-industrial) used as regressors, by data source; each dot is one unique T_{ts} value.

6 Appendix: Regression Results

Figure 5 plots per-capita wildfire $\text{PM}_{2.5}$ exposure by GMT change for the five most populous countries as well as the global average. The plots show the common estimated slope, β_c , as well as the unique fixed-effect intercepts for each model. Note that β_c is estimated for 174 countries. We also estimate β globally, which is applied to select small countries captured in the GIVE model for which our country list did not have a match. The CESM model tends to have higher exposure than other models and has the greatest variation in GMT change. In sensitivity tests we estimate β_c with data from a single model (Section 10), where the β_c results from CESM data most closely correspond to our preferred fixed-effects specification (Equation (1)).

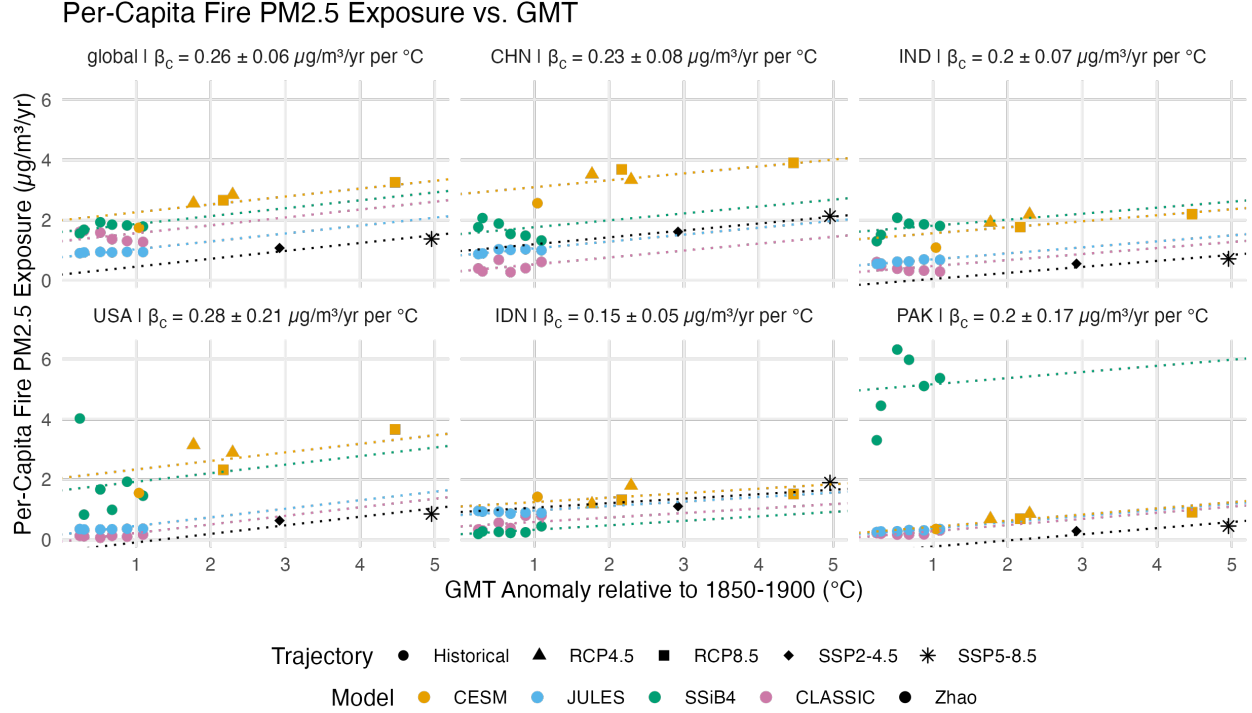


Figure 5: Per-capita wildfire PM_{2.5} exposure ($\mu\text{g}/\text{m}^3/\text{yr}$) by GMT change relative to the pre-industrial baseline ($^{\circ}\text{C}$) for the five largest population countries as well as the global average. Each plot shows 25 observations from Park et al. (2024), Pierce et al. (2017), and Zhao et al. (2025). Point color indicates wildfire model; point shape indicates warming trajectory. The five dotted lines show the fitted line for each wildfire model, using the common slope β_c and the unique fixed-effect intercept.

7 Appendix: FUND Region Decomposition

Table 1 decomposes the mean SC-CO₂ by FUND region and damage sector. The largest wildfire PM_{2.5} components are concentrated in the Former Soviet Union (\$11.29/tCO₂), United States (\$6.56), Middle East (\$5.10), and Western Europe (\$5.00)—regions combining large positive β_c estimates with substantial exposed populations. Australia and New Zealand exhibit a negative wildfire PM_{2.5} contribution (\$-0.51), reflecting negative β_c estimates for those countries (Figure 3). The largest total SC-CO₂ values are in South Asia (\$45.30), Middle East (\$41.13), and China (\$31.69), driven primarily by temperature mortality and agriculture damages.

The regional decomposition highlights a notable feature of studies applying a VSL that varies by country according to GDP per capita: Given the same physical damage (fire PM_{2.5} and temperature mortality), poorer countries will have a lower monetized damage than rich countries. This means that damages in South Asia or Sub-Saharan Africa are substantially discounted relative to those in the United States or Western Europe for mortality impacts that are comparable or even larger. The equity implications of VSL-based damage valuation are an important issue beyond the scope of this paper.

Table 1: SC-CO₂ by FUND Region (\$/tCO₂)

Fund Region	Total	wildfire PM	Temp mortality	Agriculture	Energy	Sea-level rise
Former Soviet Union	23.57	11.29	4.45	7.72	0.04	0.06
United States	22.35	6.56	13.20	1.59	0.62	0.38
Middle East	41.13	5.10	17.57	17.06	1.34	0.07
Western Europe	8.53	5.00	2.56	0.72	0.10	0.16
China	31.69	4.43	13.12	13.38	0.65	0.11
South Asia	45.30	4.11	9.95	30.08	0.99	0.16
Eastern Europe	3.82	2.15	1.25	0.38	0.01	0.02
North Africa	16.92	1.90	7.24	7.14	0.58	0.05
Canada	2.30	1.69	1.31	-0.70	-0.00	0.00
Southeast Asia	13.64	1.63	6.38	4.31	0.90	0.42
Sub-Saharan Africa	29.53	1.39	11.53	12.85	3.66	0.10
Latin America	7.35	1.36	2.06	3.35	0.37	0.22
Central America	3.94	0.60	0.86	2.28	0.13	0.06
Japan, Korea	4.13	0.23	3.26	0.52	0.04	0.09
Small Island States	0.46	0.07	0.23	0.07	0.03	0.06
Australia, New Zealand	-0.06	-0.51	1.16	-1.00	0.02	0.27
Total	254.60	47.00	96.13	99.75	9.48	2.23

8 Appendix: Qiu et al. (2026) method

8.1 Summary of Qiu et al. (2025) and Qiu et al. (2026)

Using a data-driven approach focused on the USA, Qiu et al. (2025) model the full causal chain linking GMT to wildfire emissions, emissions to smoke PM_{2.5} concentrations, and PM_{2.5} to mortality. They (i) “construct an ensemble of statistical and machine learning models that predict wildfire emissions as a function of climate and land use variables over North America”; (ii) “use surface wildfire smoke PM_{2.5} estimates derived from satellite data and machine learning models to establish an empirical relationship between wildfire emissions and smoke PM_{2.5} concentrations across the contiguous USA at a 10-km resolution”; and (iii) use a Poisson distributed-lag model to estimate the effects of annual smoke PM_{2.5} on county-level all-cause mortality rates from 2006-2019 (Qiu et al., 2025, p. 936). Qiu et al. (2026) use this framework to estimate a quadratic damage function relating GMT change to smoke-related mortality in the USA, which is incorporated into the GIVE model to compute a partial SC-CO₂. With “projections from 28 global climate models (GCMs) from the sixth phase of the Coupled Model Intercomparison Project (CMIP6),” Qiu et al. (2026, p. 2) project forward their causal model to estimate future mortality. Each of the 28 GCMs have climate variable outputs for four SSPs (shared socioeconomic pathway) and seven 20-year time periods. Additionally, they develop an empirical wildfire-fuel feedback that reduces projected wildfire emissions at higher warming levels.

8.2 Strengths and limitations of each approach

Our approach. The primary strength of our approach is global coverage. Whereas Qiu et al. (2025) require 10-km resolution smoke $\text{PM}_{2.5}$ predictions for the USA, soil moisture from NLDAS-2 (North American Land Data Assimilation System)—reportedly available only in the contiguous USA—and county-level mortality data covering the contiguous United States, the datasets we use (Park et al., 2024; Pierce et al., 2017; Zhao et al., 2025) generate globally gridded wildfire $\text{PM}_{2.5}$ concentrations that can be translated into country-level damage function parameters for all GIVE regions.

As noted in Section ??, a further advantage of our approach is that the Park et al. (2024) and Pierce et al. (2017) datasets capture dynamic feedbacks between GMT change and wildfire activity. Specifically, these process-based models predict how climate change-driven wildfires will impact vegetation growth and fuel loads, which in turn influences future wildfire activity. Qiu et al. (2026) acknowledge this as an important factor missing from their approach.

The primary limitation of our approach is limited data. Whereas Qiu et al. (2026) estimate their quadratic damage function with 784 climate simulations ($28 \text{ GCMs} \times 4 \text{ SSPs} \times 20 \text{ time periods}$), our country-level regression draws on 25 observations per country, which is sufficient to identify a linear, constant slope, but no curvature or time-temperature interactions. Additionally, our uncertainty intervals are conditional: they reflect sampling uncertainty in β_c and γ but do not fully capture disagreement across wildfire models in the GMT- $\text{PM}_{2.5}$ slope, or uncertainty arising from alternative functional forms (Section 3.4).

Qiu et al. While we rely on output from Earth System Models relating GMT to $\text{PM}_{2.5}$ concentrations, Qiu et al. (2025, p. 941) model “the full causal chain linking climate to health through wildfires (climate change-wildfire-air pollution-mortality), and [they propagate] uncertainty from each step to understand uncertainty in likely health impacts.” As they note, “one advantage of our data-driven climate-fire model over process-based model projections is its ability to isolate the factors of climate variables from other factors that influence wildfires” (Qiu et al., 2025, p. 941). In contrast, in the Pierce et al. (2017) model, land-use change assumptions (deforestation) influence future wildfire $\text{PM}_{2.5}$ concentrations—a factor for which our regression does not control—but Pierce et al. also suggest that climate variables are the primary driver of concentrations (Ford et al., 2018). Additionally, in the Qiu et al. (2025, pp. 935–936) approach, the “magnitude of smoke $\text{PM}_{2.5}$ is directly constrained by the empirical estimates of smoke $\text{PM}_{2.5}$ concentrations,” avoiding the reliance on chemical transport models that can yield estimates differing “by an order of magnitude when compared with surface observations.” Their approach also supports identification of a quadratic GMT change-mortality relationship, and Qiu et al. (2026) incorporate an empirically estimated wildfire-fuel feedback that reduces the SC- CO_2 by 68% relative to a no-feedback specification. This feedback effect is not explicitly

captured in our regression but is accounted for in our PM_{2.5} simulation data (Park et al., 2024; Pierce et al., 2017), as noted above.

The primary limitation of Qiu et al. (2026) is geographic scope. Their data requirements—particularly for satellite-derived smoke PM_{2.5} at 10-km resolution and NLDAS-2 soil moisture—seem to confine the approach to North America, preventing global application. As noted above, an additional limitation is not fully capturing climate-vegetation dynamics or lightning ignitions, both of which process-based models represent explicitly (Pierce et al., 2017).

8.3 Why Qiu et al. (2026) estimate stronger wildfire PM_{2.5} effects

Several modeling aspects of Qiu et al. (2025) and Qiu et al. (2026) result in higher excess mortality estimates at higher warming levels relative to our results (Section 9).

Climate change-PM_{2.5} concentrations relation. Qiu et al. (2026) find that 1.5°C warming increases population-weighted PM_{2.5} exposure by 0.84 $\mu\text{g}/\text{m}^3$ in the USA, while warming of 2, 3, and 4°C increases exposure by 1.2, 2.4, and 4.5 $\mu\text{g}/\text{m}^3$, respectively. By contrast, our linear regression finds that population-weighted PM_{2.5} exposure in the USA increases 0.28 $\mu\text{g}/\text{m}^3$ per 1°C warming (Section 6). Assumed constant, our estimated relation yields lower exposure at higher warming levels as well.

Smoke-specific concentration-response function. Qiu et al. (2025) empirically estimate a smoke-specific exposure-response function, which they find to have a larger impact on all-cause mortality than all-source PM_{2.5}. Their results are larger than those used in prior work—including our application of Pope et al. (2020): a 0.8% increase in all-cause mortality per $\mu\text{g}/\text{m}^3$ of *all-source* PM_{2.5} exposure—because smoke PM_{2.5} may have distinct chemical composition and toxicity. Qiu et al. (2025) state that this higher estimated exposure-response largely accounts for their higher mortality estimates relative to other studies.

Distributed-lag mortality. Qiu et al. (2025, p. 940) use a distributed-lag model and find that “exposure to annual mean smoke concentrations of 0.75-1 $\mu\text{g}/\text{m}^3$ increases the mortality rate by 1.5% in the same year, but by 2.7% ... if we also account for deaths occurring in the following year.” Qiu et al. (2026) adopt this approach, accounting for exposure in the current and prior year. Our approach applies a single-period all-source response coefficient and does not capture this multi-year mortality effect.

Quadratic damage function. Qiu et al. (2026) estimate a quadratic GMT-mortality relationship from 784 simulations spanning 0.95-6.93°C of warming. As shown in our comparison (Table 3), the quadratic function exceeds our linear function in levels above approximately 3.4°C, producing substantially more projected deaths at high warming levels (e.g., 115,667 vs. 71,494 at 6°C).

9 Appendix: USA Results

Qiu et al. (2026) estimate a partial SC-CO₂ for the U.S. accounting for wildfire PM_{2.5} mortality. This section compares our U.S. results against theirs. First, Figure 6 shows β_i estimates across the contiguous United States. Values are positive throughout nearly the entire country, with the strongest responses concentrated in the northern interior—the upper Midwest and Great Lakes region—consistent with the high β_c estimated for the United States and Canada at the country level. The weakest responses are in the Southeast and Gulf Coast. The West shows moderate-to-high values.

Grid cell estimates: Change in fire PM2.5 concentration ($\mu\text{g}/\text{m}^3$) per 1°C GMT increase

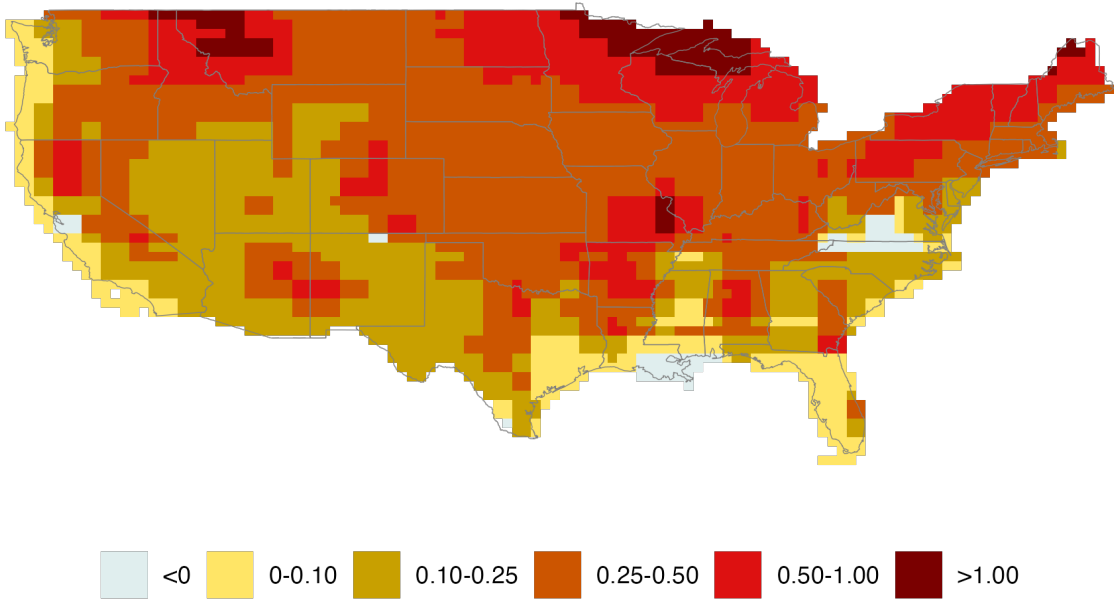


Figure 6: Grid-cell estimates of β_i , the change in wildfire PM_{2.5} concentration ($\mu\text{g}/\text{m}^3/\text{yr}$) per 1°C increase in GMT, estimated from a grid-cell-level fixed-effects regression pooling data from Park et al. (2024), Pierce et al. (2017), and Zhao et al. (2025).

The U.S. wildfire PM_{2.5} SC-CO₂ has a mean of \$6.57/ton CO₂ (95% conditional interval: \$-1.90–\$23.70) at a 2% discount rate (Figure 7, Panel A). Panel B shows that wildfire PM_{2.5} mortality represents the second-largest damage sector after temperature mortality (\$13.20) and increases the total U.S. SC-CO₂ by 41%, from \$15.83 to \$22.40. Our main U.S. result of \$6.57 is below the \$11.39 estimated by Qiu et al. (2026) (Table 2).¹⁴ Differences between our results and those of Qiu et al. (2026) are driven by different damage functions, which are discussed

¹⁴We replicate the Qiu et al. (2026) results using 100 Monte Carlo simulations rather than their published 10,000; the difference is negligible, with their published wildfire PM_{2.5} SC-CO₂ of \$11.2 closely matching our replication value of \$11.39. Minor differences in the unchanged sectors (temperature mortality, agriculture, energy, and sea-level rise) reflect the fact that incorporating an additional damage sector reduces net consumption, which alters the endogenous discount rate and thus affects all sector estimates marginally.

below along with associated excess death estimates.

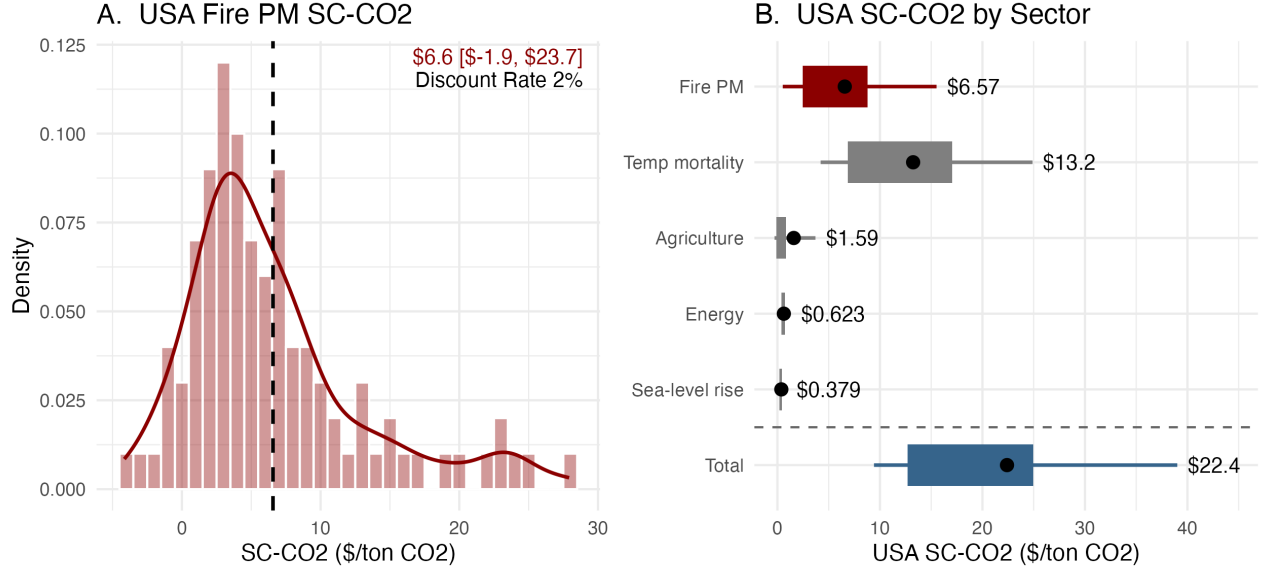


Figure 7: U.S. partial SC-CO₂ estimates at a 2% discount rate from 100 Monte Carlo simulations. (A) Distribution of the wildfire PM_{2.5} SC-CO₂ component; the dashed line marks the mean (\$6.57/ton CO₂; 95% conditional interval: \$-1.90–\$23.70). (B) SC-CO₂ by damage sector; the dot marks the mean, the box spans the 25th–75th percentile, and whiskers span the 10th–90th percentile.

Table 2: USA SC-CO₂ Comparison with Qiu et al. 2026 (\$/tCO₂)

Sector	SC-CO2	SC-CO2 Qiu et al.	Difference
Temperature-related mortality	13.24	13.17	0.07
wildfire PM _{2.5} mortality	6.57	11.39	-4.82
Agriculture	1.59	1.58	0.01
Energy	0.62	0.62	0.00
Sea-level rise	0.38	0.38	0.00
Total	22.40	27.14	-4.74

Figure 8 compares our linear damage function to the Qiu et al. (2026) quadratic function. This is illustrative only, evaluated at a fixed U.S. population of 340 million and a baseline death rate of 8.9 per 1,000. Note also that the Qiu et al. β s are averaged across their 10,000 trial runs. Table 3 reports the actual average excess deaths produced by the GIVE model—comparing our results to Qiu et al. (2026)—where population evolves over time.

Two features of Figure 8 are important. First, the dotted vertical line marks 1.7°C: the GMT change at which the *marginal* excess deaths of the Qiu et al. quadratic first exceeds that of our linear function. Recall that the *marginal* excess death is what drives wildfire PM_{2.5} SC-CO₂ estimates. Below 1.7°C, each additional degree of warming causes more *marginal* deaths under our function than under Qiu et al. (2026). On the one hand, warming of >1.7°C occurs over most years modeled in GIVE, which covers 2025 to 2300. This means that most years will have greater marginal damage from wildfire under Qiu et al. than under our results. On the other hand, the earlier decades (with less warming) when our marginal damage is greater corresponds to years with less discounting. In later years, each death is worth less in monetary terms, due to higher discounting.¹⁵ This means that, although lower, our wildfire PM_{2.5} SC-CO₂ estimates are not substantially lower than those of Qiu et al., as might be expected for the reasons outlined in Section 8.3. Second, the two curves cross in *levels* at 3.4°C, after which the Qiu et al. quadratic projects more excess deaths than our linear function.

GIVE estimates excess death results for each year, country, and iteration n . Table 3 reports U.S. annual excess deaths averaged across all year-iteration pairs where GMT change is $\pm 0.10^\circ\text{C}$ of each reported level. For example, the 1.5°C row averages over all year-iteration pairs with $T \in [1.4, 1.6]$. These results reflect the levels crossover shown in Figure 8. At 1.5°C our function projects 12,681 excess deaths annually compared to 1,128 for Qiu et al. (2026), with the gap remaining large at 2°C (18,335 and 7,172). The estimates cross in levels between 3–4°C, after which Qiu et al. project more excess deaths than our linear function at all higher warming levels (115,667 and 71,494 at 6°C).

Note that Qiu et al. (2026) express excess deaths as a function of GMT with population scaled in millions: their damage function takes the form $(\beta_1 \cdot T + \beta_2 \cdot T^2) \cdot \text{pop}_{\text{millions}}$. Our function uses the baseline mortality count M_{ct} . Additionally, we present excess wildfire PM_{2.5} deaths attributable to climate change—shown in Table 3 from Equation (2). Rather than reporting a similar result based on their quadratic damage function, which enters GIVE, Qiu et al. (2026) report *total* excess deaths from wildfire PM_{2.5}, as computed using the Qiu et al. (2025) approach. At 3°C, Qiu et al. (2026) report 64,000 total annual deaths, as compared to the 20,142 annual deaths attributable to climate change that we report in Table 3 replicating their results.

¹⁵Our discounting is identical to Qiu et al.

Table 3: Average Excess Deaths by Warming Level, Comparison with Qiu et al. 2026

Temperature	Average Excess Deaths	Average Excess Deaths Qiu et al.
1.5	12,681	1,128
2	18,335	7,172
3	23,160	20,142
4	36,228	37,611
5	40,329	60,897
6	71,494	115,667

U.S. annual excess deaths attributable to wildfire PM_{2.5} averaged across all year-iteration pairs where GMT change is $\pm 0.10^\circ\text{C}$ of the reported warming level.

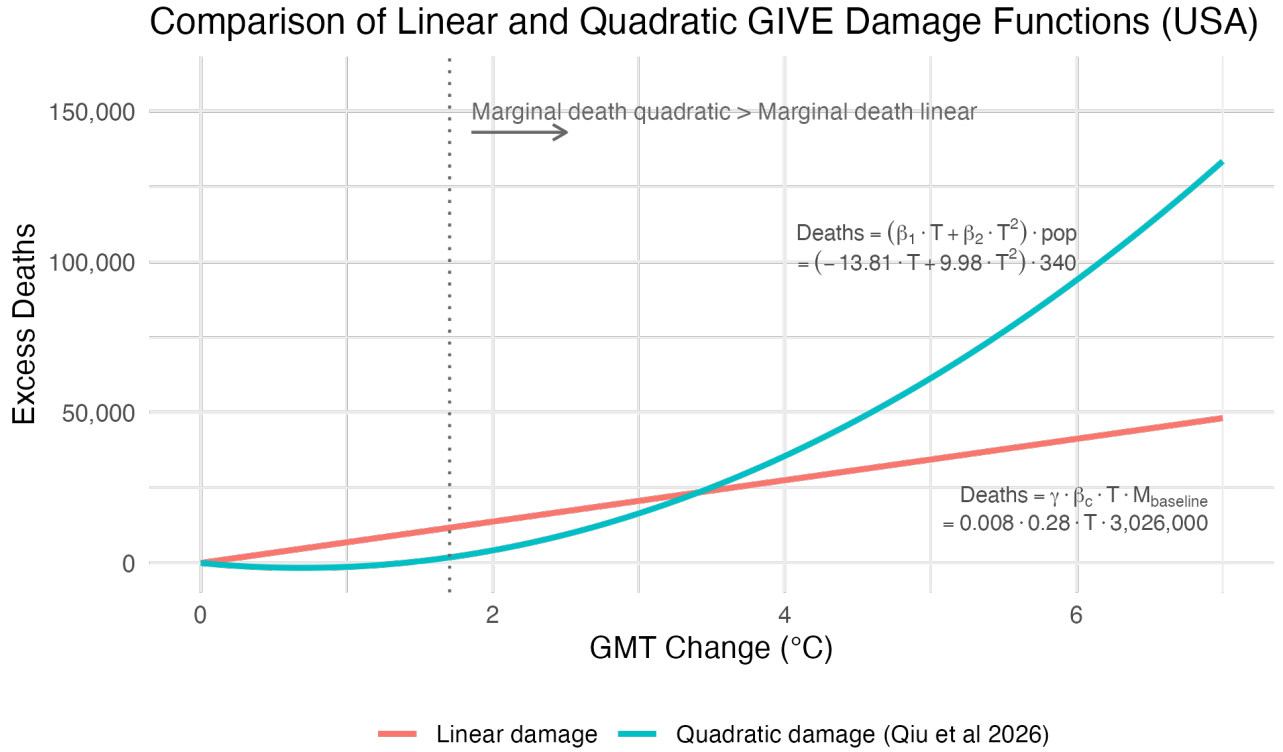


Figure 8: Excess wildfire PM_{2.5} mortality in the USA as a function of GMT change, comparing a linear damage function (this paper) and the Qiu et al. (2026) quadratic function. The dotted vertical line marks the GMT change at which the marginal excess death of the quadratic function exceeds that of the linear (1.7°C). Both functions are evaluated at a fixed U.S. population of 340 million and baseline death rate of 8.9 per 1,000. The β s for the Qiu et al. (2026) quadratic are representative—averaging over 10,000 values—not the exact parameters entering GIVE. The linear β_c is our estimated coefficient for the USA.

10 Appendix: Sensitivity Test - Single Model Regressions

Here we present sensitivity tests where β_c is estimated using observations from a single wildfire PM_{2.5} simulation model, rather than pooling all models. We have:

$$P\bar{M}_{cts} = \beta_c T_{ts} + \alpha_c + \varepsilon_{cts}, \quad (3)$$

where $P\bar{M}_{cts}$ and T_{ts} are identical variables as used before. The coefficient of interest, β_c has the same interpretation as in Equation (1). However, Equation (3) no longer has model fixed-effects and α_c is a country-specific intercept.

We estimate Equation (3) using data from each wildfire model except Zhao et al. (2025), for which the two observations per country are insufficient to estimate standard errors¹⁶. For each individual model and for the preferred fixed-effects specification, Table 4 shows regression summary statistics: the percentage of countries with positive β_c and p-values < 0.05, the minimum, 25th percentile, median, 75th percentile, and maximum. The CESM results most closely match the fixed-effect specification results: they share similar variation in estimated β_c and a similar proportion of countries with a positive GMT-fire PM_{2.5} relationship. This suggests that the CESM observations—which have the widest temperature variation and tend to have higher exposure (Section 5)—are influential in determining the pooled slope in the fixed-effects specification. Relative to the fixed-effects model with more observations, the estimate precision drops off in the single-model specifications (higher p-values). Note also the wider variation among Park et al. (2024) models and their disagreement, with CLASSIC showing only 43% of countries with a positive GMT-fire PM_{2.5} relationship.

Table 4: Summary statistics of country-level β_c estimates across regression specifications.

Model	% $\beta_c > 0$	% $p < .05$	Min	P25	Median	P75	Max
FE	86	60	-1.05	0.07	0.18	0.39	3.33
CESM	88	45	-0.70	0.10	0.25	0.52	3.41
CLASSIC	43	43	-12.02	-0.33	-0.05	0.08	9.40
JULES	58	42	-4.77	-0.06	0.01	0.07	2.79
SSiB4	81	21	-7.03	0.01	0.14	0.82	25.58

Across regression specifications, Table 5 shows β_c and p-value results for the top nine most populous countries—which have a larger impact on SC-CO₂ results—and the global average. Only Nigeria has a negative β_c in the fixed-effects specification, while all countries have a positive relation for CESM. CLASSIC is the only model showing a negative relation for the global

¹⁶With only two observations from Zhao et al. (2025), there are zero residual degrees of freedom.

average. Reflected in both Table 4 and Table 5, the single-model specifications—particularly the models from Park—appear to show implausible outliers. Outliers include Nigeria within the CLASSIC model and the U.S. within SSiB4. With fewer observations, single data points with outlier $\text{PM}_{2.5}$ exposure values can dominate the slope estimate, as illustrated for the U.S. under SSiB4 in Figure 5. In contrast, the fixed-effects specification mitigates this by pooling across models—increasing observations and thereby reducing the influence of any single outlier—while also absorbing level differences among models such that the slope is identified from within-model variation.

Table 5: β_c estimates and p-values across specifications for largest countries

Country	FE		CESM		CLASSIC		JULES		SSiB4	
	β	p	β	p	β	p	β	p	β	p
global	0.262	0.001	0.381	0.046	-0.475	0.004	0.035	0.122	0.223	0.260
CHN	0.229	0.009	0.310	0.123	0.184	0.481	0.159	0.094	-0.741	0.021
IND	0.199	0.009	0.257	0.165	-0.322	0.021	0.173	0.004	0.505	0.218
USA	0.284	0.190	0.509	0.100	0.042	0.425	0.022	0.106	-1.262	0.483
IDN	0.148	0.014	0.051	0.640	0.598	0.020	-0.074	0.097	0.190	0.122
PAK	0.204	0.250	0.136	0.103	0.074	0.365	0.102	0.004	1.718	0.291
NGA	-0.188	0.716	0.162	0.343	-8.758	0.001	0.117	0.433	2.599	0.097
BRA	0.166	0.131	0.273	0.298	0.004	0.994	-0.267	0.131	-0.236	0.605
RUS	1.735	0.002	2.612	0.135	0.143	0.041	-0.029	0.216	0.130	0.082
MEX	0.145	0.035	0.217	0.210	-0.371	0.115	-0.000	0.995	0.147	0.619

References

- Anthoff, D. (2026). MimiSSPs.jl. <https://github.com/anthofflab/MimiSSPs.jl>
- Bayham, J., Yoder, J. K., Champ, P. A., & Calkin, D. E. (2022). The Economics of Wildfire in the United States [eprint: <https://doi.org/10.1146/annurev-resource-111920-014804>]. *Annual Review of Resource Economics*, 14(1), 379–401. <https://doi.org/10.1146/annurev-resource-111920-014804>
- Ford, B., Val Martin, M., Zelasky, S. E., Fischer, E. V., Anenberg, S. C., Heald, C. L., & Pierce, J. R. (2018). Future Fire Impacts on Smoke Concentrations, Visibility, and Health in the Contiguous United States [eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1029/2018GH000144>]. *GeoHealth*, 2(8), 229–247. <https://doi.org/10.1029/2018GH000144>
- Orellano, P., Kasdagli, M.-I., Pérez Velasco, R., & Samoli, E. (2024). Long-Term Exposure to Particulate Matter and Mortality: An Update of the WHO Global Air Quality Guidelines Systematic Review and Meta-Analysis. *International Journal of Public Health*, 69, 1607683. <https://doi.org/10.3389/ijph.2024.1607683>
- OWID. (2026). Global average temperature anomaly relative to 1861-1890. <https://ourworldindata.org/grapher/temperature-anomaly?v=1&csvType=full&useColumnShortNames=false>
- Park, C. Y., Takahashi, K., Fujimori, S., Jansakoo, T., Burton, C., Huang, H., Kou-Giesbrecht, S., Reyer, C. P. O., Mengel, M., Burke, E., Li, F., Hantson, S., Takakura, J., Lee, D. K., & Hasegawa, T. (2024). Attributing human mortality from fire PM_{2.5} to climate change. *Nature Climate Change*, 14(11), 1193–1200. <https://doi.org/10.1038/s41558-024-02149-1>
- Pierce, J., Val Martin, M., & Heald, C. (2017). *Estimating the Effects of Changing Climate on Fires and Consequences for U.S. Air Quality, Using a Set of Global and Regional Climate Models* (tech. rep.). Joint Fire Science Program. <https://www.frames.gov/catalog/25211>
- Pope, C. A., Coleman, N., Pond, Z. A., & Burnett, R. T. (2020). Fine particulate air pollution and human mortality: 25+ years of cohort studies. *Environmental Research*, 183, 108924. <https://doi.org/10.1016/j.envres.2019.108924>
- Qiu, M., Callahan, C. W., Higuera-Mendieta, I., Rennels, L., Parthum, B., Diffenbaugh, N. S., & Burke, M. (2026). Valuing wildfire smoke-related mortality benefits from climate mitigation. *Proceedings of the National Academy of Sciences*, 123(8), e2533772123. <https://doi.org/10.1073/pnas.2533772123>
- Qiu, M., Li, J., Gould, C. F., Jing, R., Kelp, M., Childs, M. L., Wen, J., Xie, Y., Lin, M., Kiang, M. V., Heft-Neal, S., Diffenbaugh, N. S., & Burke, M. (2025). Wildfire smoke exposure and mortality burden in the USA under climate change. *Nature*, 647(8091), 935–943. <https://doi.org/10.1038/s41586-025-09611-w>

- Rennert, K., Errickson, F., Prest, B. C., Rennels, L., Newell, R. G., Pizer, W., Kingdon, C., Wingenroth, J., Cooke, R., Parthum, B., Smith, D., Cromar, K., Diaz, D., Moore, F. C., Müller, U. K., Plevin, R. J., Raftery, A. E., Ševčíková, H., Sheets, H., ... Anthoff, D. (2022). Comprehensive evidence implies a higher social cost of CO₂. *Nature*, *610*(7933), 687–692. <https://doi.org/10.1038/s41586-022-05224-9>
- Wimberly, M. C., Wanyama, D., Doughty, R., Peiro, H., & Crowell, S. (2024). Increasing Fire Activity in African Tropical Forests Is Associated With Deforestation and Climate Change [eprint: <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2023GL106240>]. *Geophysical Research Letters*, *51*(9), e2023GL106240. <https://doi.org/10.1029/2023GL106240>
- Zhao, J., Zheng, B., Ciais, P., Chen, Y., Gasser, T., Canadell, J. G., Zhang, L., & Zhang, Q. (2025). Global warming amplifies wildfire health burden and reshapes inequality. *Nature*, 1–3. <https://doi.org/10.1038/s41586-025-09612-9>